

Functional Specification

Status legend: **Implemented** · **Partial** · **Stub** / **Planned**

1. Authentication & session

Capability	Status	Notes
Staff login (username/password)	Implemented	JWT returned
Current user profile (/auth/me)	Implemented	
Change password	Implemented	Min 8 chars; confirm match
Logout (API + audit)	Implemented	POST /auth/logout records AUTH LOGOUT
Remember-me (local vs session storage)	Implemented	Frontend
Failed-attempt lockout	Implemented	LoginAttemptService
Role-based route guards	Implemented	Angular authGuard / portalGuard

2. Dashboard

Capability	Status	Notes
Summary counts (customers, accounts, employees)	Implemented	
Today's transaction count	Partial	Hardcoded 0 until Transaction module exists

3. Customers

Capability	Status	Notes
Search / paginate customers	Implemented	
View customer detail	Implemented	
Create customer	Implemented	Optional first account on create
Update customer	Implemented	
Soft/hard delete	Implemented	ADMIN only for DELETE
Open additional accounts from detail	Implemented	Current (policy-aware opening deposit); Loan button present

4. Accounts

Capability	Status	Notes
Open account	Implemented	Policy-validated opening deposit
List / search accounts	Implemented	<code>/app/accounts</code>
List accounts by customer	Implemented	
Get account by id	Implemented	<code>GET /accounts/{id}</code>
Update account status	Implemented	API present
Deposit (post-open)	Implemented	Credits balances + writes CREDIT ledger row; Kafka email to customer
Withdraw	Implemented	DEBIT + Kafka email
Transfer	Implemented	DEBIT+CREDIT; Kafka email to destination customer
Account detail + ledger UI	Implemented	Account detail + paged transactions
List pagination	Implemented	First/Last + Go to page (<code>list-pagination</code>)
Account policies (create + get active)	Implemented	Seeded INR policies; no LOAN seed
Opening-deposit dialog (Current)	Implemented	Uses <code>GET /account-policies</code>
Kafka customer email	Implemented	See MODULES/Notification.md

5. Employees (users)

Capability	Status	Notes
List / create / update employees	Implemented	ADMIN; multi-role assignment (<code>roleNames</code>)
Role CRUD + access codes	Implemented	<code>/roles</code> ; union of role authorities at login
Account policies UI	Implemented	Management card; ADMIN/MANAGER edit

6. Branch

Capability	Status	Notes
Branch master data	Stub	Only <code>branchCode</code> string (often <code>0001</code>)

7. Loans

Capability	Status	Notes
Loan product / EMI / collateral	Stub	
<code>AccountType.LOAN</code> account opening	Partial	Enum + UI affordance; needs policy; not a loan domain

8. Transactions & reports

Capability	Status	Notes
Transaction ledger entity/API	Implemented	Write on deposit / withdraw / transfer / opening; staff <code>/transactions</code> UI; account-detail UI; dashboard today-count still stub
Reports (charts + PDF)	Implemented	Trends, account mix, approvals; OpenPDF
Beneficiaries	Implemented	Portal CRUD

9. Customer portal & approvals

Capability	Status	Notes
Portal accounts / txn list	Implemented	<code>/portal/*</code>
Portal transfer	Implemented	Immediate or pending approval
Transfer approval workflow	Implemented	Staff queue + my history
Transfer approval threshold on customer	Implemented	

10. Notifications

Capability	Status	Notes
Kafka bank-action events	Implemented	Topic <code>bankone.notifications</code>
Customer email on open / deposit / withdraw / transfer	Implemented	Recipient = <code>customers.email</code>
Local Mailpit / Render SendGrid HTTPS	Implemented	See MODULES/Notification.md (<code>APP_KAFKA_ENABLED</code>)

11. Audit

Capability	Status	Notes
JPA created/updated by fields	Partial	<code>AuditableEntity</code> on some entities
Activity event trail API / UI	Implemented	Categories incl. AUTH login/logout
Transfer approval audit history	Implemented	
Historical backfill	Implemented	Admin; idempotent

Non-functional (current)

Area	Behavior
AuthN	JWT HMAC, 1h expiry (<code>jwt.expiration</code>)
AuthZ	Role + <code>ACCESS_*</code> authorities (<code>AppAccess</code> / <code>SecurityConfig</code>)
CORS	localhost, bankone.local, 192.168.0.4
Logging	Hibernate SQL DEBUG (dev-oriented)
Deploy	Open Liberty WAR primary path; Render Docker + Aiven + SendGrid
Messaging	Kafka (optional via <code>APP_KAFKA_ENABLED</code>)
Learning roadmap	See <code>TECH_LEARNING_PLAN.md</code> (rate limit + cache local implemented ; outbox, ...)

Deferred hardening items are tracked in `.cursor/rules/deferred-hardening.mdc` . Platform learning topics (not the same as hardening) are tracked in [TECH_LEARNING_PLAN.md](#). Redis cache-aside how-it-works: [MODULES/Caching.md](#).